

Pranesh S

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SUMMARY

Agentic AI / AI/ML Engineer (Generative AI) and B.Tech (AI & ML) graduate — **Immediate Joiner** — building production LLM systems: agentic pipelines on **MCP** and **AWS Bedrock AgentCore**, RAG platforms, and **7B-parameter model fine-tuning**. Open-source maintainer: **100+ public repos**, **270+ stars**, **3 PyPI packages**, **35+ live Hugging Face Spaces**.

TECHNICAL SKILLS

Generative AI & Agents: LLM fine-tuning (LoRA, QLoRA, Unsloth), RAG, embeddings & semantic search, quantization (4-bit, GGUF), AI Agents, Agentic AI, Multi-Agent Systems, MCP, LangChain, LangGraph, CrewAI, PageIndex, prompt engineering, LLM red teaming

Machine Learning & Deep Learning: Transformers, PyTorch, Deep Learning, NLP, computer vision (EfficientNetV2, MTCNN), model evaluation

Languages & Backend: Python, Java, SQL, Data Structures & Algorithms, Flask, FastAPI, REST APIs, PostgreSQL, Redis, ChromaDB

Cloud & MLOps: AWS Bedrock AgentCore, Azure AI Foundry, Modal, Docker, Hugging Face, PyPI, Git/GitHub (Version Control), Vercel, Render, Supabase

EXPERIENCE

AI Engineering Intern — Harman DTS

Feb 2026 – Jun 2026

- Developed an **NL2SQL2NL** system on the **Model Context Protocol (MCP)** enabling context-aware, multi-turn natural-language querying over relational databases (**SQL** generation + result summarization) for business users.
- Extended the pipeline with **Matplotlib-based visualization tools**, letting non-technical stakeholders generate business-case charts directly from conversational queries.
- Deployed an **Agent** to **AWS Bedrock AgentCore**, enabling automated testing of LLM applications in a managed cloud agent runtime.
- Built a **Multi-Agent Workflow Generator** that composes multi-agent AI workflows directly from client requirements, cutting manual orchestration design effort.

PROJECTS

RagObserve — RAG Observability Platform | *Python, FastAPI, SQLite, PyPI* github.com/Pranesh-2005/RagObserve

- Built a **local-first observability platform for RAG applications** — tracing retrievals, LLM calls, latency, and cost across providers — with built-in **LangChain** and **LlamaIndex** integrations, running fully offline with zero external dependencies.

Deepfake Video Detector (27★ on GitHub) | *Python, EfficientNetV2, MTCNN, Gradio* hf.co/spaces/PraneshJs/fakevideodetect

- Designed an open-source deepfake detection system flagging AI-generated video via frame-by-frame analysis, combining **MTCNN** face detection with **EfficientNetV2** classification; shipped a live **Gradio** demo on Hugging Face with a user feedback loop for model improvement.

PromptGuard — Prompt-Injection Detection Model | *Python, BERT, Transformers, PyPI* hf.co/PraneshJs/PromptGuard

- Trained a **BERT text-classification model** on prompt-injection datasets to guard LLM apps against malicious inputs; published to Hugging Face (**150+ downloads**) and shipped as the **guardix** PyPI library.

Persona Mimic Chatbot — Qwen2.5-7B + LoRA + RAG Memory | *Python, Unsloth, Modal, GGUF, Gradio*

- Engineered a data pipeline converting **120K+ raw WhatsApp messages** into **14K instruction-tuning samples** (parsing, dedup, burst-windowing); fine-tuned **Qwen2.5-7B-Instruct** with **LoRA (Unsloth)** on serverless **Modal A10G** GPUs.
- Added long-term memory via **RAG**: embedded 30K+ chat bursts with multilingual-e5 for top-k retrieval at chat time; exported **4-bit GGUF quantized** models for local inference and deployed a Gradio Space on Hugging Face.

ACHIEVEMENTS & OPEN SOURCE

- github-readme-stats-fast**: performance-focused GitHub profile stats generator — **270+ stars**, **59 forks**, plus a companion GitHub Action
- PyPI packages**: *guardix* (prompt-injection guard), *Text-Emotion* (text emotion detection), *DocTok* (document token counter)
- 35+ live Hugging Face Spaces** spanning RAG, agents, model-internals visualizers (GPT-2, ViT, YOLO), and AI tooling

EDUCATION

Bannari Amman Institute of Technology

2022 – 2026

B.Tech in Artificial Intelligence and Machine Learning — CGPA: 7.97/10

CERTIFICATIONS

- Anthropic**: Introduction to Model Context Protocol · Model Context Protocol: Advanced Topics
- Microsoft Applied Skills**: IDP with Azure AI Document Intelligence · NLP solutions with Azure AI Language
- freeCodeCamp**: Machine Learning with Python